

Statement of intent

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PhD position, auction-based support schemes for renewable energy
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I am applying for the doctoral position on auction-based support schemes for renewable energy in Europe. This statement sets out the training I would bring to it and the training I would have to acquire.

Background in energy

My energy training is at Cologne, where the specialisation is taught by the Institute of Energy Economics. I took *Energy Markets and Regulation* with Jun.-Prof. Dr. Oliver Ruhnau, whose chair is in energy market design, and finished it with 1.3 on a scale where 1.0 is the maximum, followed by *Growth, Energy and Climate Change* with PD Dr. Dietmar Lindenberger.

My final seminar paper, written this summer in the Panel Data Methods seminar at Cologne, is on carbon leakage from the EU Emissions Trading System. It is a research design rather than a set of estimates. I wrote it as the groundwork for a master's thesis on the topic that I decided in the end not to write, and what it contains is the specification, the treatment definition, the data sources and the identifying assumptions paired with the threats each one answers.

The question is whether increases in the effective EU ETS carbon cost raised EU imports of emission-intensive goods from Morocco, Algeria, Egypt and Tunisia between 2005 and 2021, relative both to less emission-intensive goods and to comparison partners. Aggregate studies of EU manufacturing trade find no statistically significant leakage effect (Naegele and Zaklan, 2019). My argument is that an aggregate null can conceal leakage concentrated in corridors where regulatory asymmetry is large, transport costs are low, and emission-intensive capacity already exists.

The specification is a triple difference: exposure varies across products through sectoral emission intensity, over time through the effective carbon cost, and across partners through corridor membership, with product-specific, partner-specific and bilateral shocks absorbed by three sets of fixed effects. One feature of the treatment definition is why I am writing to this project rather than to a general applied-micro post. I define the effective carbon cost as the allowance price attenuated by the sector-year free-allocation share, so a sector receiving full free allocation faces an effective cost of zero whatever the headline EUA price does. The objection is that tradable allowances carry the permit price as marginal opportunity cost, so a lump-sum grant should leave the output margin untouched. EU allocation is not lump-sum in that sense: activity thresholds, closure provisions and partial-cessation rules tie allowances to continued production. The headline price of a climate instrument and the incentive it creates are different objects, and the gap between them is a property of the rules.

That is the same problem an auction for renewable capacity presents. An award price is not the support a project effectively receives once pre-qualification requirements, penalties, completion bonds and local content conditions are priced into the bid. I have already written one paper about measuring the effective stringency of a European climate instrument rather than its nominal one, and I would like the next four years to be about the other one.

Proposed contribution

Of the four directions in the call, the two I would take up are pre-qualification requirements together with completion bonds, and local content requirements, and my reason is the identification problem they present. Auction design in Europe is not a single policy. Member states adopted auctions at different dates and revised the rules at different dates again: penalty levels, bid bonds, realisation deadlines, technology-specific against technology-neutral lots, and local content conditions have each been introduced and amended on separate national timetables. This is staggered adoption of a treatment whose intensity varies across adopters, and it is the setting in which conventional panel estimators fail quietly.

My master's thesis, graded 1.0, was built for that setting. Using the estimator of Callaway and Sant'Anna (2021) on a panel of roughly five million linked employer-employee records that I constructed, the doubly-robust estimate is a wage effect of +3.04 per cent in low-exposure regions, against -0.65 per cent from two-way fixed effects on the same data. The sign reverses for the reason Goodman-Bacon (2021) identifies, and I know it does because I ran the decomposition.

Auction theory predicts how pre-qualification and penalty regimes change bidding behaviour and completion rates. Testing those predictions against the European record needs an estimator that survives staggered adoption with heterogeneous effects, and the diagnostics that establish whether it has. That is the part of the project I could do on arrival.

Training I would have to acquire

The call asks for graduate training in microeconomic theory, industrial organization and game theory. My master's microeconomics grade is 3.0, which is average on the German scale, and the transcripts accompanying this application show it.

The fuller picture is more mixed than that one mark. At bachelor level I took microeconomics with a grade of 1.0 and advanced microeconomics with Prof. Dr. Axel Ockenfels, on strategic thinking and competition, with a grade of 1.3. My specialisation module in current topics in energy and environmental policy was also 1.3. So game theory and competition are not new to me, and I learned them from someone whose own field is market and auction design.

What I would genuinely be building is graduate industrial organization at the level of Mas-Colell, Whinston and Green and of Osborne and Rubinstein. That is coursework and reading, and a first year is designed to accommodate it. The empirical craft is the harder thing to add later, and I have that already.

Practicalities

I complete my M.Sc. in Economics at Cologne this month and am available from October 2026, which fits the autumn start. I hold IELTS Academic 8.0, and I speak German and Tunisian Arabic natively and French at B1. I hold EU citizenship, so no work permit is required.

My referees are Prof. Dr. Pia Pinger, who chaired my master's thesis, and Jun.-Prof. Dr. Oliver Ruhnau, who taught and examined me in energy market design. Contact details appear on the CV. The thesis and the seminar paper on the EU ETS are available on request.

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